Yafei Zhou

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Affiliation: School of Advanced Engineering, Great Bay University, China
Excepted Graduation: Jun. 2027 | Personal Website: https://yafeizhou.me/

Education

Great Bay University & Shenzhen University (Joint Training Program) — **Sep. 2024 – Jun. 2027 (Excepted)**
M.Eng. in Control Engineering
Relevant Coursework: Control Engineering, Modern Control Theory, Advanced Engineering Mathematics, Automatic Control Systems Practice, General English, Technical English.
GPA: 3.2/5.0

Dongguan University of Technology — **Sep. 2020 – Jun. 2024**
B.Eng. in Intelligent Manufacturing Engineering
Relevant Coursework: Advanced Mathematics, Linear Algebra, Digital Electronics, Analog Electronics, C Programming, C++ Programming and Application Development, Sensor and Measurement Technology, NX Fundamentals and Advanced Applications.
GPA: 3.6/5.0

Research Interests

Vision-Based Tactile Sensors (VBTS), Robotic Tactile Perception, Contact Force Estimation, Tactile Robotic Manipulation, and Multimodal Perception for Embodied Intelligence

Research Experience

Robot Perception with Vision-Based Tactile Sensors — **Sep. 2024 – Present**
Research Intern, Daimon Robotics

- Developed a **visuotactile sensor data acquisition pipeline**, including tactile image capture, six-axis force/torque measurements, and timestamp recording and synchronization across multiple sensors.
- Analyzed surface contact states of visuotactile sensors using **optical-flow features extracted from tactile images**, and modeled the relationship between tactile deformation and contact forces.
- Conducted **tactile force regression experiments using CNN- and ResNet-based models**, and evaluated model performance across different sizes and configurations of visuotactile sensors.
- Integrated visuotactile perception into **robotic inspection and manipulation tasks**, enabling stable grasping, object interaction, and other contact-rich operations based on tactile feedback.

Tactile Reasoning for Multimodal Large Models — **Dec. 2025 – Jun. 2026**
Research Intern, Daimon Robotics

- Conducted research on **tactile reasoning for multimodal large models**, investigating how visuotactile signals can assist models in reasoning about physical properties such as object hardness, surface roughness, and contact deformation, while mitigating conflicts between visual and tactile information.
- Designed a **handheld visuotactile–force data acquisition system**, incorporating IMU-based pose estimation and gravity compensation for a six-axis force/torque sensor to obtain accurate ground-truth contact forces.Conducted **tactile force regression experiments using CNN- and ResNet-based models**, and evaluated model performance across different sizes and configurations of visuotactile sensors.
- Participated in the collection, organization, and preprocessing of **large-scale visuotactile datasets**, including tactile image acquisition, timestamp synchronization across visuotactile sensors, six-axis force/torque sensors, IMUs, and RealSense D405 cameras, as well as cross-sensor data alignment and quality inspection.
- Designed **tactile reasoning evaluation tasks** to assess multimodal models in tactile perception, physical consistency reasoning, and vision–touch conflict reasoning.

Manuscripts Under Review

- Yingxin Lai*, **Yafei Zhou***, Fucai Zhu, Siyu Zhu, and Weihao Yuan. *Touch-R1: Reinforcing Touch Reasoning in MLLMs.* arXiv preprint arXiv:2605.27154, 2026. Under review at NeurIPS 2026.

Skills

- **Programming**: Python, C
- **English Proficiency**: CET-6
- **Deep Learning**: PyTorch, CNNs, ResNet, Optical-Flow-Based Representation Learning
- **Robotics & Hardware**: Hands-on experience with vision-based tactile sensors (DM-Tac, GelSight, Xense, Tac3D), robotic manipulators (KUKA LBR iiwa 7 R800, Kinova Gen3), six-axis force/torque sensors (Kunwei, Xinjingcheng), Intel RealSense D405 RGB-D cameras, and IMUs
- **Software & Tools**: Git, Linux, OpenCV, Docker

Honors & Awards

- **Spark New Talent Award,** *Daimon Robotics Technology Co., Ltd.*, 2026
- **Outstanding Graduate,** *DGUT*, 2024
- **National Inspirational Scholarship,** *DGUT*, 2021 & 2022
- **Outstanding Student Scholarship,** *DGUT*, 2022
- **Second-Class Academic Scholarship,** *DGUT*, 2022
- **Enterprise Scholarship,** *DGUT*, 2021
- **Outstanding Student,** *DGUT*, 2021